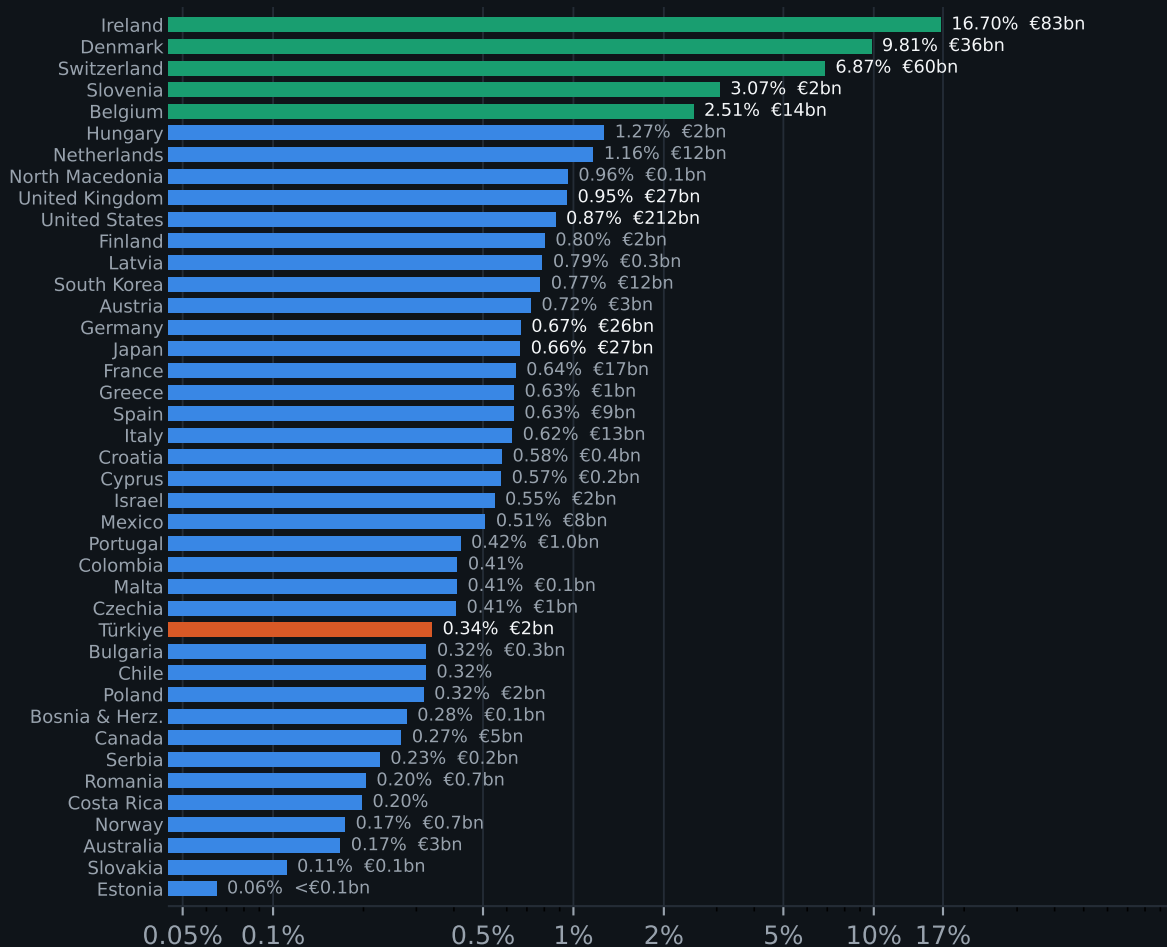


Pharmaceutical manufacturing in the national economy

41 countries, each at its most recent published year.



Share of total value added, log scale. Label: share, then size in euro.

Green: above 2% of the economy. Türkiye in orange.

Three findings

01

Scale is remarkably uniform.

In every large economy, pharmaceutical manufacturing accounts for approximately 1% of output — and has done so for three decades.

02

Concentration is the real variable.

Shares of 7–17% occur only in small economies where a single industry has come to dominate: Ireland, Denmark and Switzerland.

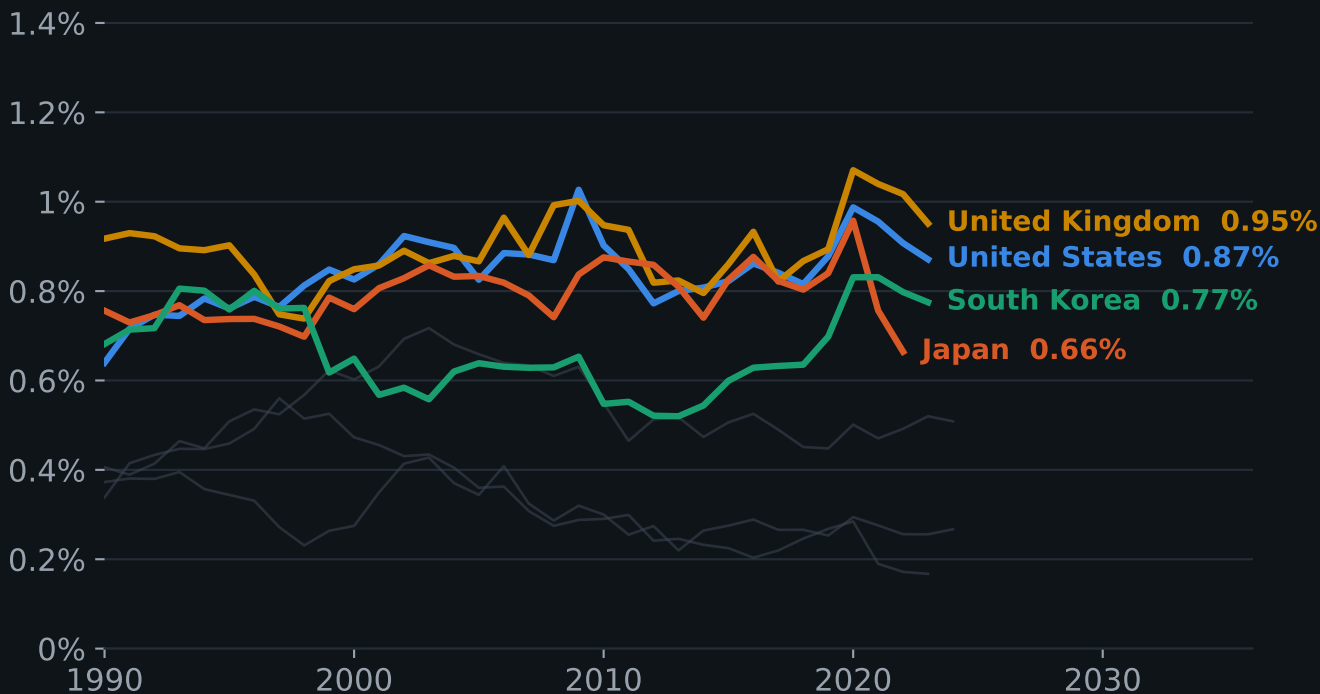
03

Composition differs more than scale.

Belgium exports vaccines, Ireland the active substance behind GLP-1 medicines, Germany antibodies and plasma — distinctions invisible in the headline economic figures.

Large economies hold pharmaceuticals near 1%

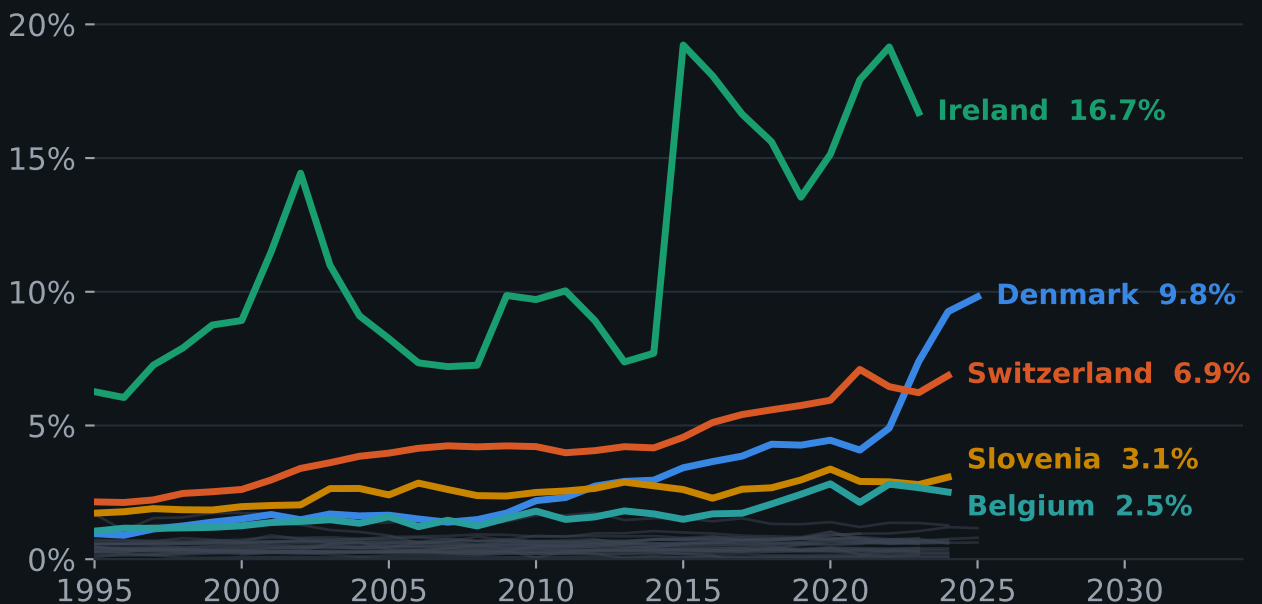
Share of total value added, 1990 to most recent year.



The band has held through the genomics era, the shift to biologics and the pandemic. Grey lines: Canada, Mexico, Australia, France, Italy, Spain, Germany.

The exceptions are all small economies

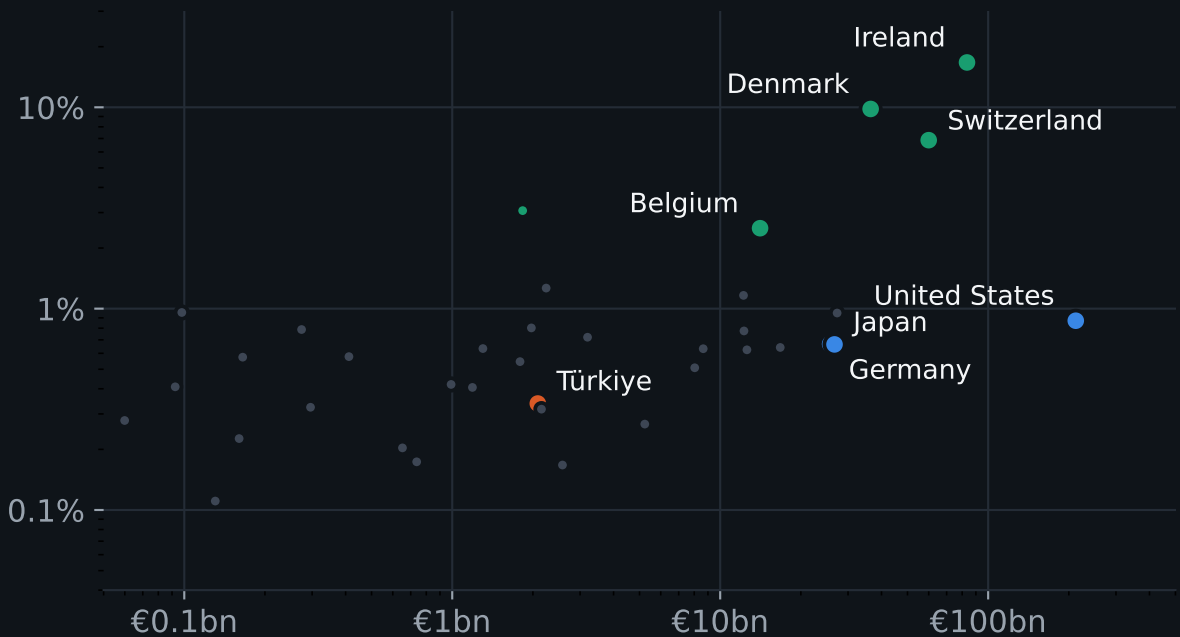
Pharmaceutical share of the economy, 1995-2025.



In 2023 Germany and Denmark each produced roughly €25bn of pharmaceutical value added. That is 0.67% of the German economy and 7.4% of the Danish one. Concentration, not sector size, determines macroeconomic exposure. Ireland's step up in 2015 is a national-accounts restatement, not a jump in output.

Producing a lot and depending on it are different things

Pharmaceutical value added, € billion, against share of the economy.



€212bn

United States output,
0.87% of its economy

€83bn

Ireland output,
16.7% of its economy

4

countries in both
corners — all small

Two routes to concentration

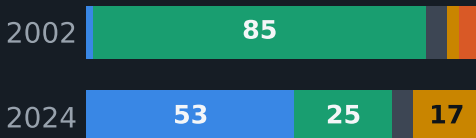
Ireland repositioned its output. Denmark deepened a single franchise. Both reached scale; the composition differs.

Ireland

6.3% → 16.7%

of value added, 1995–2023

Exports €15bn → €99bn



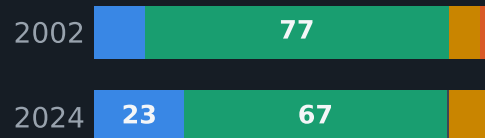
Export composition, %

Denmark

1.0% → 9.8%

of value added, 1995–2025

Exports €4bn → €22bn



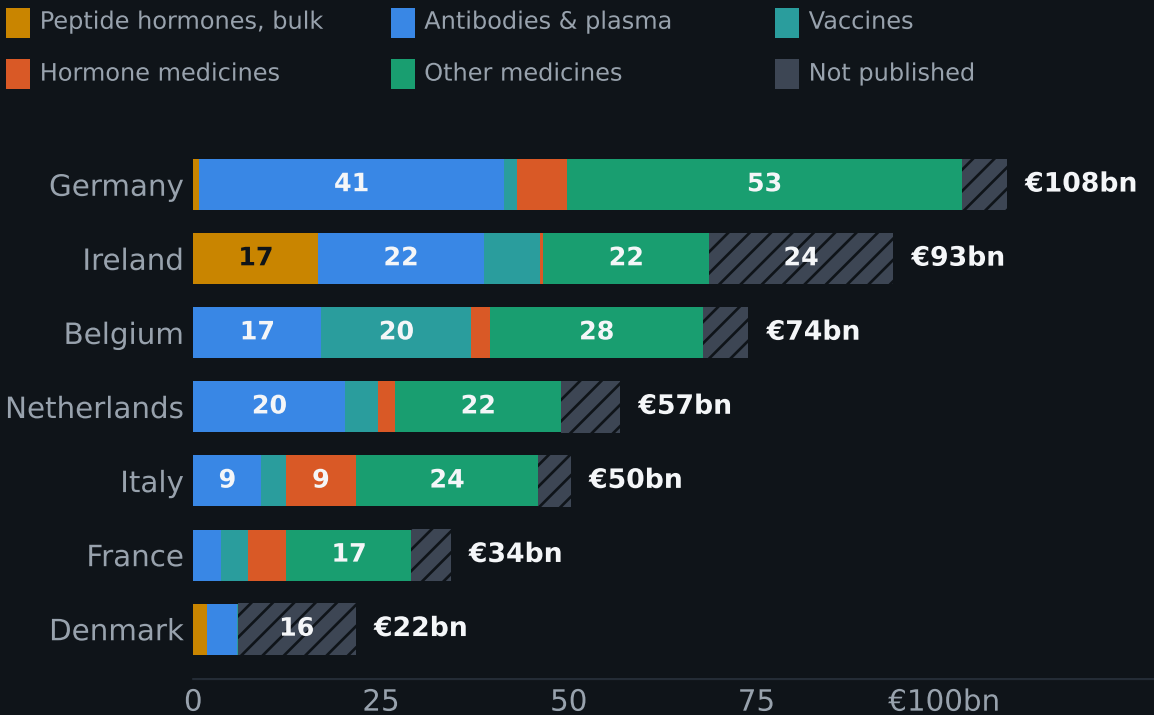
Export composition, %

■ Biologics / vaccines ■ Finished dose ■ Hormones

Ireland moved from 85% finished dose in 2002 to 53% biologics and 17% hormones in 2024 — a change of product, not only of volume. Denmark stayed dose-led, scaling one franchise. Irish shares step up in 2015 with the national-accounts restatement; the export mix is unaffected by it.

What they actually make

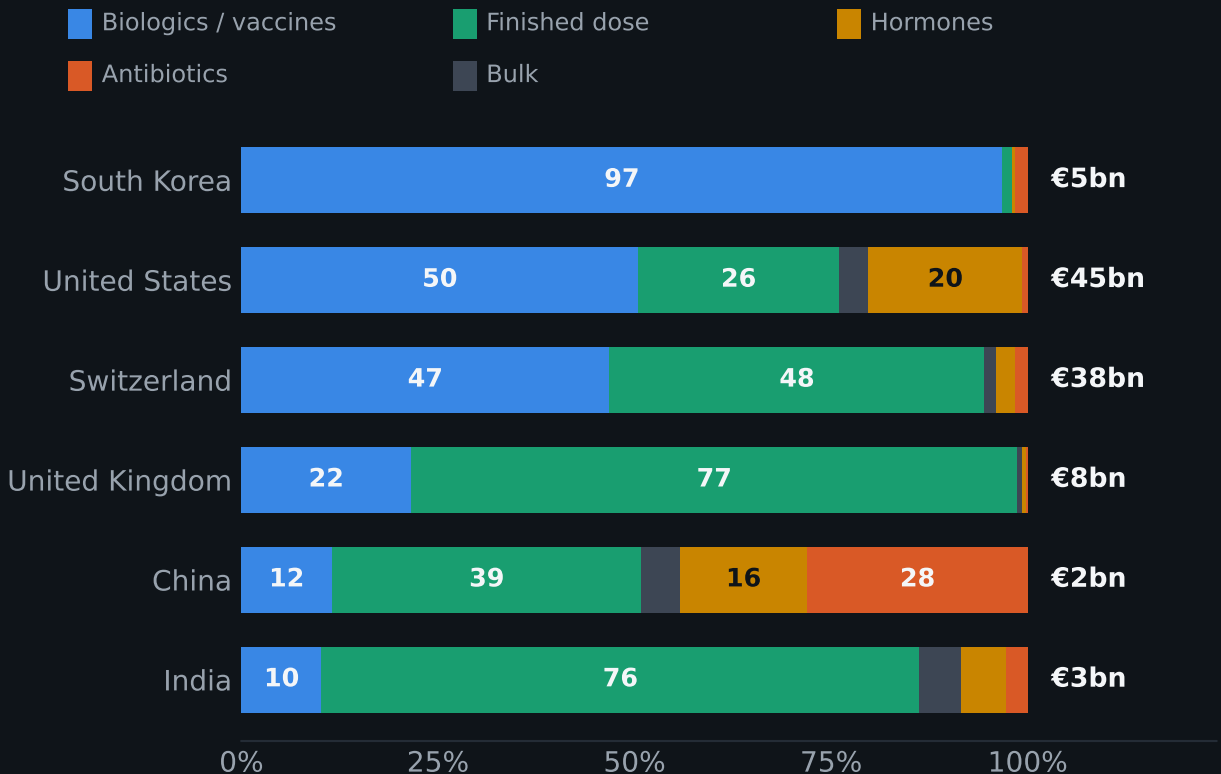
Exports to world by therapeutic class, 2024, € billion.



Belgium is Europe's vaccine plant (€20bn). Ireland's largest single class is now bulk peptide hormones — the GLP-1 and insulin-analogue active substance. Germany spreads across antibodies, plasma and small molecules. Denmark publishes almost none of its detail.

Switzerland is invisible in EU export data

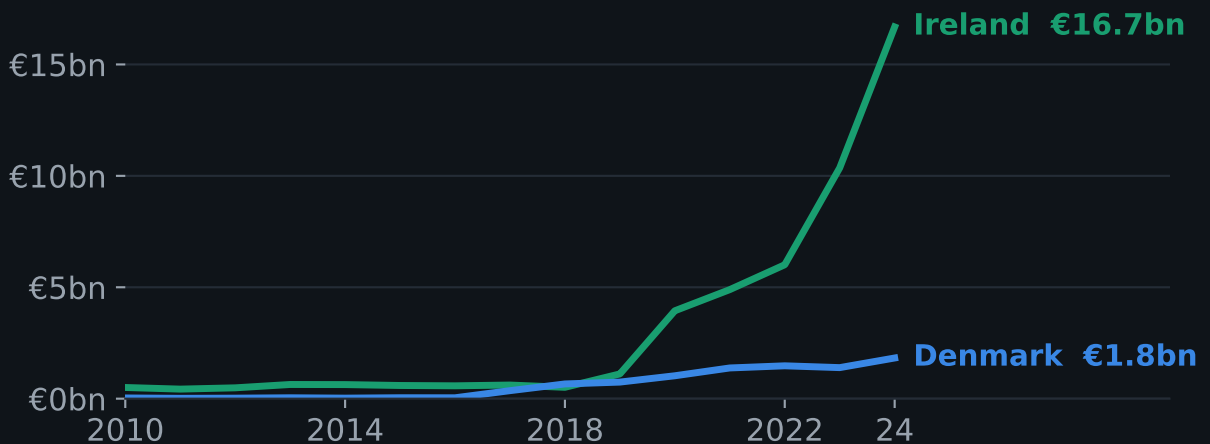
EU customs records only what EU members report. Non-EU producers show up solely as what the EU buys from them, 2024.



Switzerland ships €38bn into the EU, split almost evenly between biologics and finished medicines. Korea ships little else: 97% biologics, built in under a decade. A partial view — it misses everything these countries sell outside the EU.

One class explains Ireland's decade

Bulk peptide hormones (HS 2937.19) — the active substance behind GLP-1 and insulin-analogue medicines. Exports to world, € billion.



€0.5bn in 2018 to €16.7bn in 2024 — a thirty-fold increase in six years, and now Ireland's single largest pharmaceutical export class.

47%

of Irish pharma exports go to the United States

€43bn

Belgian vaccine exports at the 2022 peak

14%

of Danish exports have a named destination

Denmark withholds its insulin, hormone and other-medicament codes and most partner detail: too few firms to publish without disclosing them.

What the analysis supports

Economic analysis

Danish, Irish and Swiss aggregates warrant an ex-pharmaceutical view before macro conclusions are drawn.

Industrial strategy

Composition can be changed within a decade — Ireland and South Korea repositioned output rather than expanding it.

Türkiye

0.34% of the economy, broadly unchanged since 2003 against a 0.5% median. The constraint is positioning, not data.

Method · NACE/ISIC C21 value added, Eurostat and OECD STAN, in national currency. Composition from HS chapter-30 trade, a proxy for output. Full sources and caveats online.

namikakmandev.github.io/pharma-gdp-share.html